

tasks such as:

Traffic management: Advanced routing and load balancing between microservices.

Security: Mutual TLS (mTLS) for encrypted communication and access controls.

Observability: Monitoring and tracing for performance insights.

▪ **How it works**

The service mesh consists of:

Data plane: Lightweight proxies (sidecars) deployed within each pod, managing traffic and collecting telemetry data.

Control plane: Centralised management for configuring proxies, issuing TLS certificates, and enforcing policies.

This setup abstracts network management from application code, allowing for secure, reliable, and efficient service communication.

▪ **Key features**

Traffic management: Supports advanced routing and deployment strategies.

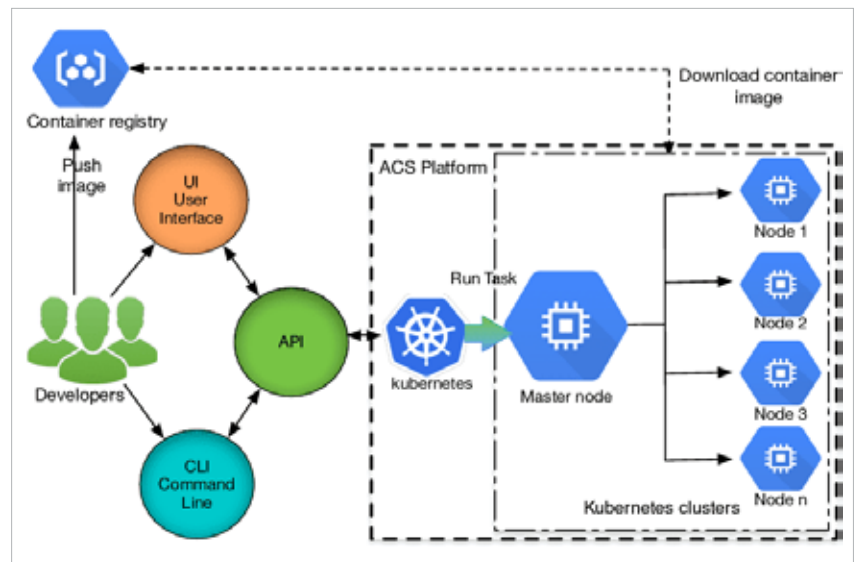


Figure 1: Kubernetes orchestration overview

Security: Automates encryption and access control.

Observability: Provides detailed monitoring and tracing.

Integrates well with observability tools.

Linkerd: Known for simplicity and performance, it offers essential features like mTLS and basic traffic management.

An example is VMware NSX Advanced Load Balancer, which integrates with Tanzu Service Mesh

▪ **Popular tools**

Istio: Provides extensive features like Layer 7 load balancing and mTLS.

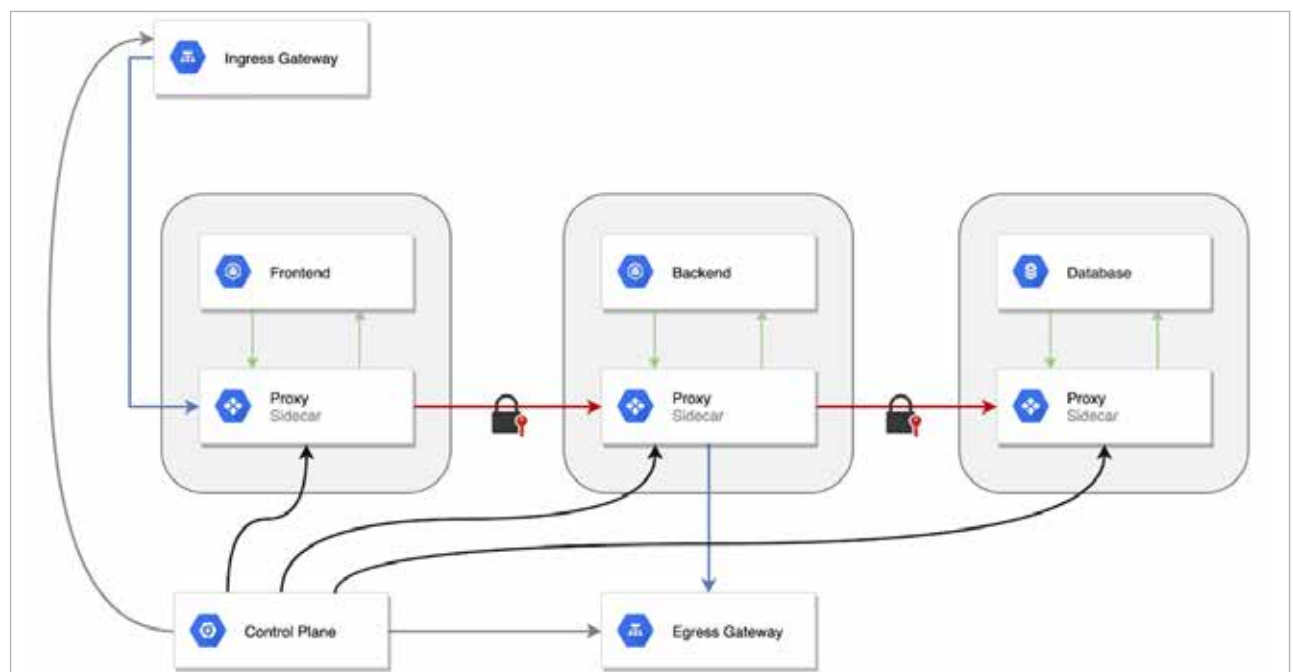


Figure 2: Architecture of Kubernetes service mesh